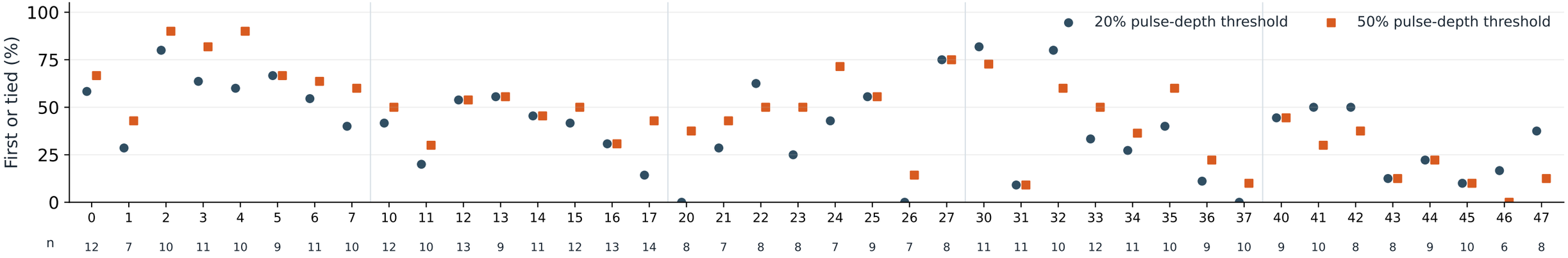
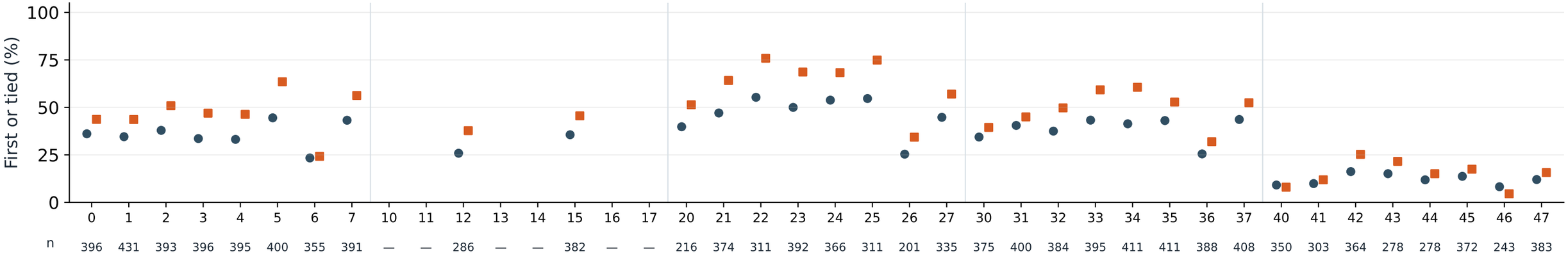
No measured channel always starts first in either APA

First selected negative pulse before −100 µs. Count a channel as tied for first when it is within one 16 ns sample.

APA1: 15 timed pulses · tied leaders in 12/15 at 20% and 14/15 at 50%



APA2: 490 timed pulses · tied leaders in 272/490 at 20% and 404/490 at 50%



Horizontal labels are channel numbers. n is the number of pulses where that channel can be timed. Missing channels are not late channels.

APA1: channel 30 is first or tied in 9/11 pulses at 20%. Channels 2 and 4 lead or tie in 9/10 at 50%. None always leads.

Compare the same channels at both thresholds, with at least 8 timed channels per event. Many channels can tie in one pulse.

The timing is not corrected for channel or cable delays. A recurring measured lead would not by itself identify a physical source.